Buvana Seshathri

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Education

University of Texas at Dallas — Dallas, TX
Master of Science in Computer Science, Intelligent Systems – GPA: 3.88/4.0 — *Aug 2024 – May 2026*

Sri Sairam Engineering College — Chennai, India
Bachelor of Engineering in Computer Science – GPA: 9.41/10 — *Aug 2019 – May 2023*

Experience

HPC Student Assistant – HPC Facilitation — Jun 2025 – May 2026
University of Texas at Dallas — *Dallas, TX*

- Built a Python CLI with a prompt-based interface that lets a non-technical research team run 10+ Stable Diffusion models on HPC clusters without writing SLURM job scripts, turning manual GPU job setup into a self-serve workflow.
- Translated a research team's requirements into documented, reusable GPU workflows, cutting onboarding time by 50%.
- Supported 500+ researchers across 15 departments as primary HPC contact for Linux, access and SLURM scheduling.
- **Technologies:** Python, PyTorch, Hugging Face, CUDA, SLURM, Linux, Bash

Member of Technical Staff, Java Developer — Jul 2022 – Jul 2024
Zoho Corporation — *Chennai, India*

- Led root-cause analysis of critical production issues on ServiceDesk Plus, Zoho's flagship ITSM product, including a severe outage where inefficient database access on a 1M+ row table exhausted the connection pool and crashed the server.
- Rebuilt the customer-facing Errorlog module end-to-end on a REST API model, designing new database tables, building the APIs from scratch and redesigning the UI with filters, cutting page load time by 99.9% (5 to 10 minutes to 200 ms).
- Owned SSL/TLS certificate infrastructure, JRE upgrades and the Errorlog module as sole Directly Responsible Individual (DRI), handling every ticket, feature, hotfix and security patch across live, customer-facing production systems.
- Diagnosed and fixed issues live in enterprise customer environments over remote sessions, from server startup failures to production errors, explaining each root cause and fix directly to the customer's IT team.
- Resolved 50+ high-priority customer-reported production incidents in on-call rotation, closing 90%+ within SLA.
- **Technologies:** Java, JavaScript, SQL, MySQL, PostgreSQL, MSSQL, REST APIs, Microservices, Redis, Tomcat, Maven

Projects

ScholarMind – RAG Research Assistant | *Python, FastAPI, LangChain, FAISS, Groq*

- Built a retrieval-augmented generation (RAG) pipeline that splits uploaded research PDFs into 800-character overlapping chunks with LangChain, embeds them locally with MiniLM and indexes each paper in its own FAISS vector store.
- Served answers from Llama 3.1 8B on Groq through a FastAPI backend using only the top 4 retrieved passages, returning page-level citations with source snippets and saving per-paper conversation history in SQLite.

Multi-LLM Chat – Classifier-Based Model Router | *Python, FastAPI, scikit-learn, OpenAI, Gemini, Groq*

- Built a multi-session chat app that labels each message as writing, code, research or general with a TF-IDF + logistic regression classifier and routes it to OpenAI, Gemini or Groq via a user-editable task-to-model mapping.
- Built a fallback chain that retries a failed call on another provider, then a local Llama 3.2 model via Ollama, behind one shared client interface, and stores multi-session conversation history in SQLite.

Telly – Explainable Recommender, Published Chrome Extension | *Embeddings, Recommender Systems, LLM APIs*

- Built a hybrid recommender that ranks titles by MiniLM-L6-v2 embedding similarity, facet lift and TMDB neighbour links with MMR diversity re-ranking, evaluated offline by hit-rate and MRR on a held-out 20% of watch history.
- Added opt-in bring-your-own-key LLM re-ranking across five providers (OpenAI, Anthropic, Gemini, Groq, OpenRouter) that sees only a taste summary and about 20 candidates, with guardrails that drop invented titles.

Technical Skills

Languages: Python, Java, C, SQL
AI/ML: PyTorch, TensorFlow/Keras, scikit-learn, Hugging Face, Pandas, NumPy, Fine-Tuning (LoRA)
Generative AI: RAG, LangChain, FAISS, Vector Embeddings, Prompt Engineering, OpenAI and Gemini APIs
Backend & Data: FastAPI, Flask, REST APIs, PostgreSQL, Kafka, Redis
Infrastructure: AWS, Docker, GitHub Actions, CUDA, SLURM

Research

- **Cross-Modal Transfer Learning**, Perceptive Mobile AI and Mobile Computing Lab, UTD (2026): transferring gesture recognition learned on Wi-Fi signals to GNSS satellite signals, studying generalization across sensing modalities.
- **IEEE Publication**, I2CT 2023: *Comparison of VGG-19 and ResNet-50 Algorithms in Brain Tumor Detection.*